

A Practical Guide to Vision transformer

TOPIC -- VISION TRANSFORMER

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V$$

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ViT-VQGAN In ViT-VQGAN, there are two ViT encoders and a discriminator. One encodes 8x8 patches of an image into a list of vectors, one for each patch. The vectors can only come from a discrete set of "codebook", as in vector quantization. Another encodes the quantized vectors back to image patches. The training objective attempts to make the reconstruction image (the output image) faithful to the input image. The discriminator (usually a convolutional network, but other networks are allowed) attempts to decide if an image is an original real image, or a reconstructed image by the ViT. The idea is essentially the same as vector quantized variational autoencoder (VQVAE) plus generative adversarial network (GAN). After such a ViT-VQGAN is trained, it can be used to code an arbitrary image into a list of symbols, and code an arbitrary list of symbols into an image. The list of symbols can be used to train into a standard autoregressive transformer (like GPT), for autoregressively generating an image. Further, one can take a list of caption-image pairs, convert the images into strings of symbols, and train a standard GPT-style transformer. Then at test time, one can just give an image caption, and have it autoregressively generate the image. This is the structure of Google Parti.

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DINO Like the Masked Autoencoder, the DINO (self-distillation with no labels) method is a way to train a ViT by self-supervision. DINO is a form of teacher-student self-distillation. In DINO, the student is the model itself, and the teacher is an exponential average of the student's past states. The method is similar to previous works like momentum contrast and bootstrap your own latent (BYOL). The loss function used in DINO is the cross-entropy loss between the output of the teacher network ($f_{\theta_t'}$) and the output of the student network (f_{θ_t}). The teacher network is an exponentially decaying average of the student network's past parameters: $\theta_t' = \alpha \theta_t + (1 - \alpha) \theta_{t-1}$.

$\theta_t + \alpha (1 - \alpha) \theta_{t-1} + \dots$. The inputs to the networks are two different crops of the same image, represented as $T(x)$ and $T'(x)$, where x is the original image. The loss function is written as $L(f_{\theta_t}(T(x)), f_{\theta_t}(T'(x)))$. One issue is that the network can "collapse" by always outputting the same value (y), regardless of the input. To prevent this collapse, DINO employs two strategies:

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The Masked Autoencoder took inspiration from denoising autoencoders and context encoders. It has two ViTs put end-to-end. The first one ("encoder") takes in image patches with positional encoding, and outputs vectors representing each patch. The second one (called "decoder", even though it is still an encoder-only Transformer) takes in vectors with positional encoding and outputs image patches again.

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A space attention layer is a self-attention layer where each query patch $q_{x,y,t}$ attends to only the key and value patches $k_{x',y',t'}$ and $v_{x',y',t'}$ such that $t = t'$. A time attention layer is where the requirement is $x' = x$, $y' = y$ instead. The TimeSformer also considered other attention layer designs, such as the "height attention layer" where the requirement is $x' = x$, $t' = t$. However, they found empirically that the best design interleaves one space attention layer and one time attention layer.

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Comparison with CNNs Typically, ViT uses patch sizes larger than standard CNN kernels (3x3 to 7x7). ViT is more sensitive to the choice of the optimizer, hyperparameters, and network depth. Preprocessing with a layer of smaller-size, overlapping (stride < size) convolutional filters helps with performance and stability. This different behavior seems to derive from the different inductive biases they possess:

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Sharpening: The teacher network's output is sharpened using a softmax function with a lower temperature. This makes the teacher more "confident" in its predictions, forcing the student to learn more meaningful representations to match the teacher's sharpened output. Centering: The teacher network's output is centered by averaging it with its previous outputs. This prevents the teacher

from becoming biased towards any particular output value, encouraging the student to learn a more diverse set of features. In January 2024, Meta AI Research released an updated version called DINOv2 with improvements in architecture, loss function, and optimization technique. It was trained on a larger and more diverse dataset. The features learned by DINOv2 were more transferable, meaning it had better performance in downstream tasks. In August 2025, Meta AI Research released DINOv3, an update to DINOv2. It introduced image-text alignment like CLIP. It scaled up the model to 7B parameters and the training dataset to 1.7B images (obtained by diversity-sampling an initial dataset with 17B images). Architecturally, it introduced two improvements: Gram anchoring and axial RoPE (Rotary Positional Embeddings) with jittering. Gram anchoring applies teacher-student self-distillation for the Gram matrix between the feature vectors of the patches of an image. It avoids the previously observed problem of degradation of dense feature maps: While performance on global tasks (like classification) continued to improve, performance on dense tasks (like segmentation) would peak early and then decline, with feature maps becoming noisy. Axial RoPE makes the model more robust to varying image resolutions, scales, and aspect ratios.

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LayerNorm immediately after each attention and feedforward layer ("res-post-norm"); scaled cosine attention to replace the original dot product attention; log-spaced continuous relative position bias, which allows transfer learning across different window resolutions.

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Instead of performing self-attention over the entire sequence of tokens, one for each patch, it performs "shifted window based" self-attention, which means only performing attention over square-shaped blocks of patches. One block of patches is analogous to the receptive field of one convolution. After every few attention blocks, there is a "merge layer", which merges neighboring 2x2 tokens into a single token. This is analogous to pooling (by 2x2 convolution kernels, with stride 2). Merging means concatenation followed by multiplication with a matrix. It is improved by Swin Transformer V2, which modifies upon the ViT by a different attention mechanism: